

Guideline Title: Magnesium Imbalance

Guideline Number: GUID-3203-M005

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** To recognize the patient with a magnesium imbalance and to provide and maintain adequate oxygenation/ventilation and tissue perfusion. To provide emergent treatment, correct the imbalance and provide expeditious transport to the nearest appropriate medical facility.
- II. **DEFINITIONS:** None
- III. **DEPARTMENT GUIDELINE:**
 - a. Establish care as defined by GUID3203-G002 – *General Patient Care*.
 - b. **Hypomagnesemia:**
 - i. Treatment depends on the degree of deficiency and the clinical effects. Oral replacement is appropriate for mild symptoms, while IV replacement is indicated for severe clinical effects. Discuss with sending or receiving physician regarding treatment.
 - ii. If Magnesium level confirmed < 1.5 or patient is symptomatic (neuromuscular irritability, hyperactive deep tendon reflexes, muscle cramps, muscle fibrillation, Trousseau and Chvostek signs, dysarthria, and dysphagia from esophageal dysmotility, CNS hyperexcitability, irritability, combativeness, disorientation, psychosis, ataxia, vertigo, nystagmus), consider **Magnesium Sulfate 1 gram IV** over 30 minutes.
 - iii. If cardiac dysrhythmias or seizures are present, administer **Magnesium Sulfate 1-2 grams IV push over 2 minutes** Treat seizures with benzodiazepines per GUID3203-M015 - *Seizures*.
 - c. **Hypermagnesemia:**
 - i. Calcium can be effective in reversing the physical findings associated with hypermagnesemia such as nausea, vomiting, skin flushing, weakness, lightheadedness, depressed levels of consciousness, disappearance of deep tendon reflexes, hypotension, vasodilation, dysrhythmias, including atrial fibrillation, and intraventricular conduction delay. Use caution if patient is on Digitalis. For symptomatic Adult patients:
 1. [PREFERRED] **Calcium Gluconate 1-2 grams IV** over 3 min IV slowly from sending facility.
 - a. (If the patient is on digitalis, more caution is required as calcium can increase the toxic effects of digitalis. Add the appropriate dose to 100ml NS and infuse over 20-30 minutes.) – **OR** –
 - b. **Calcium Chloride 1 gram IV** slowly over 10 minutes, preferentially administered

Guideline Title: Magnesium Imbalance

Guideline Number: GUID-3203-M005

through central line or in the setting of acute decompensation/ cardiac arrest.
Monitor for local irritation at the injection site.

IV. **DOCUMENTATION:** EMS Charts

V. **REFERENCES:**